

Chapter 2:
Lesson 2-1:
How Breathing Works

Breathing helps us take in oxygen and release carbon dioxide.



A. Choose the Correct Answer

Choose the correct answer.



1. **Which statement best describes breathing?**
   a) Taking in carbon dioxide only
   b) Taking in oxygen and releasing carbon dioxide
   c) Taking in food and releasing oxygen
   d) Releasing oxygen and taking in carbon dioxide

2. **Which gas is present in a [illegible] exhaled air than in room air?**
   a) Oxygen
   b) Nitrogen
   c) Carbon dioxide
   d) Hydrogen

3. **Which material is used in the activity to compare the carbon dioxide in exhaled air and room air?**
   a) Salt water **b) Limewater** c) Distilled water d) Oil

4. **A student wants to compare the amounts of oxygen and carbon dioxide in two air samples. Which tool should be used?**
   a) Scissors **b) Gas-detection tubes** c) Safety goggles d) Plastic bags only

5. **Which observation provides evidence that exhaled air contains more carbon dioxide than room air?**
   a) Limewater turns milky more quickly with exhaled air.
   b) Exhaled air becomes blue.
   c) Room air turns limewater red.
   d) Exhaled air does not react with limewater.



B. Write True or False

Think Like a Scientist!

Write True or False.

1. Respiration is one of the important life processes carried out by living organisms. (**True**)
2. During inhalation, air containing oxygen enters our lungs. (**True**)
3. During exhalation, carbon dioxide leaves our lungs. (**True**)
4. The two plastic bags in the experiment contain air collected from exactly the same source. (**False**)
5. Safety goggles should be worn when carrying out the limewater experiment. (**True**)



Small Scientists Make a Big Difference!

Healthy Lungs Brighter Tomorrows!



Explore Learn Grow!

Chapter 2:
Lesson 2-1:
How Breathing Works



Healthy Lungs Happy You!



Small Steps Big Scientists!

Breathing helps us take in oxygen and release carbon dioxide.

C. Write the Scientific Term

Use what you have learned!

1	The process of taking air into the lungs.	Inhalation
2	The process of pushing air out of the lungs.	Exhalation
3	The gas needed by the body to release energy from food.	Oxygen
4	The gas released from the body during breathing.	Carbon dioxide
5	Small tubes containing chemicals that can be used to test the amounts of particular gases.	Gas-detection tubes



Breathe Learn Be Healthy!

D. Give Reasons for the Following

Think Like a Scientist!

1. Our body needs a continuous supply of oxygen.
   Because oxygen is needed to release energy from food (during respiration).
2. Two plastic bags are used in the breathing experiment instead of only one.
   To compare the exhaled air with normal room air.
3. One plastic bag is filled with exhaled air while the other is filled with normal room air.
   To test and compare the amount of carbon dioxide in the two air samples.
4. The two bags should be shaken after adding the same amount of limewater.
   To mix the air with the limewater so it can react.
5. Exhaled air and room air give different results when tested with gas-detection tubes.
   Because exhaled air contains more carbon dioxide than room air.

E. What Happens When...?

Predict Think Explain!

1. A person breathes in?
   Air enters the lungs through the nose or mouth.
2. A person breathes out?
   Air leaves the lungs through the nose or mouth.
3. Limewater is added to a bag of exhaled air and the bag is shaken?
   Limewater turns milky (cloudy).
4. The oxygen gas-detection tube is used to compare exhaled air with room air?
   The oxygen level is lower in exhaled air than in room air.
5. The carbon dioxide gas-detection tube is used to compare exhaled air with room air?
   The carbon dioxide level is higher in exhaled air than in room air.





Clean Air Healthy Lungs Brighter Futures!

Chapter 2 Respiration

Lesson 2-2:

How Do Lungs Work?



Breathe Learn Grow!

Try

1 (1) What gas do we take into our body when we breathe in? **Oxygen**

(2) When comparing the air we breathe in and the air we breathe out, which gas increases in the air we breathe out? Choose the correct answer from below.

A. Oxygen
B. Carbon dioxide
C. Neither increases

Clean Air Healthy Lungs Happier You!

2 (1) What is the organ in our chest called that exchanges oxygen and carbon dioxide? **The lungs**

(2) Fill in the blanks with the correct words to complete the sentence.

Through the (① **blood vessels (or capillaries)** around the lungs, (② **oxygen** in the air is taken into the blood, and (③ **carbon dioxide** in the blood is let out into the air.

Exercise

1 (1) What happens to limewater when you blow breathed-out air into it? Choose the correct answer from below.

A. It turns milky white.
B. It turns blue.
C. It disappears.

Science in Everyday Life!

(2) Which statements correctly explain breathing? Choose all correct answers from below.

A. The air we breathe out has no oxygen at all.
B. The air we breathe out also has oxygen.
C. The air we breathe out has more oxygen than the air we breathe in.
D. The air we breathe out has more carbon dioxide than the air we breathe in.

Breathe Well Live Well Be Amazing!

2 (1) Which statement correctly explains gas exchange in the lungs? Choose the correct answer from below.

A. It lets out oxygen from the blood into the air.
B. It takes in oxygen from the air into the blood.
C. It takes in carbon dioxide from the air into the blood.

(2) What carries the oxygen taken in by the lungs to the whole body? Choose the correct answer from below.

A. Blood
B. Muscles
C. Bones





Explore Learn Grow!

Chapter 2

Lesson 2-2:

How Do Lungs Work?

Our lungs help us take in oxygen and remove carbon dioxide.



Healthy Lungs Happy You!

A. Choose the Correct Answer

Choose the correct answer.

1 Which structure is the main entrance for air into the respiratory system?

a) Alveoli **b) Nose** c) Diaphragm d) Bronchioles

2 Which sequence correctly shows the path of air through the respiratory system?

a) Nose → bronchi → trachea → alveoli

b) Nose → trachea → bronchi → bronchioles → alveoli

c) Nose → alveoli → bronchi → trachea

d) Trachea → nose → bronchioles → alveoli

3 Where does gas exchange between the air and blood take place?

a) Trachea b) Nose **c) Alveoli** d) Diaphragm

4 During inhalation, which change helps air enter the lungs?

a) The chest cavity becomes smaller.

b) The diaphragm moves upward.

c) The chest cavity becomes larger.

d) The lungs become smaller.

5 A fish obtains the oxygen it needs mainly through its:

a) lungs. **b) gills.** c) nose. d) bronchi.

Breathe In Oxygen Breathe Out Carbon Dioxide



Your Lungs Work Hard for You!

B. Write True or False

Write True or False.

Think Check Be a Scientist!

1. The nose helps filter, warm, and moisten the air we breathe in. (**True**)
2. The trachea carries air from the nose or mouth toward the lungs. (**True**)
3. Bronchioles are larger tubes than the bronchi. (**False**)
4. Cats breathe air using lungs. (**True**)
5. Oxygen absorbed by a cat's lungs can be transported by the blood to different parts of its body. (**True**)

Healthy Lungs Brighter Futures!

